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Section: 07

Answer to the Question No-1

My student id: 24101226 , $a=6, b=2, c=2$

$A=7, B=3, C=3$

(a) $f(x) = 7x^2 + 3x + 3$

(b) Interval is $[0, 2]$

$$\Delta x = \frac{2-0}{n} = \frac{2}{n}$$

Using mid-point approximation.

$$x_k^* = x_{k-\frac{1}{2}} = a + (k-\tfrac{1}{2})\Delta x = (k-\tfrac{1}{2})\frac{2}{n}$$

$$f(x_k^*) = \cancel{7x_k^2}\ 7\left\{(k-\tfrac{1}{2})\frac{2}{n}\right\}^2 + 3.(k-\tfrac{1}{2})\frac{2}{n} + 3$$

$$= 7.4\,(k^2 - k + \tfrac{1}{4})\frac{1}{n^2} + 6\,(k-\tfrac{1}{2})\frac{1}{n} + 3$$

$$= 28\left(\frac{k^2}{n^2} - \frac{k}{n^2} + \frac{1}{4n^2}\right) + 6\left(\frac{k}{n} - \frac{1}{2n}\right) + 3$$

$$f(x_k^*)\Delta x = \left\{28\left(\frac{k^2}{n^2} - \frac{k}{n^2} + \frac{1}{4n^2}\right) + 6\left(\frac{k}{n} - \frac{1}{2n}\right) + 3\right\}\frac{2}{n}$$

$$= 56\frac{k^2}{n^3} - 56\frac{k}{n^3} + 56\cdot\frac{1}{4n^3} + 12\frac{k}{n^2} - 12\frac{1}{2n^2} + 6\frac{1}{n}$$

$$\text{Area} = \lim_{n\to\infty}\sum_{k=1}^{n} f(x_k^*)\Delta x$$

$$= \lim_{n\to\infty}\sum_{k=1}^{n}\left(56\frac{k^2}{n^3} - 56\frac{k}{n^3} + 14\cdot\frac{1}{n^3} + 12\cdot\frac{k}{n^2} - 6\cdot\frac{1}{n^2} + 6\cdot\frac{1}{n}\right)$$

$$= 56\cdot\frac{1}{3} - 0 + 0 + 12\cdot\frac{1}{2} - 0 + 6$$

$$= \frac{92}{3}$$

(Answer)

Answer to the Question No-2

Yes, it is possible to evaluate $\int_{-\infty}^{\infty} \frac{dx}{e^x+e^{-x}}$ using improper integral.

let. $I = \int_{-\infty}^{\infty} \frac{dx}{\frac{(e^x)^2+1}{e^x}} = \int \frac{e^x dx}{1+(e^x)^2} = \tan^{-1} e^x + C$

$$\int_{-\infty}^{\infty} \frac{dx}{e^x+e^{-x}} = \int_{-\infty}^{c} \frac{dx}{e^x+e^{-x}} + \int_{c}^{\infty} \frac{dx}{e^x+e^{-x}}$$

$$= \lim_{a\to-\infty} \int_{a}^{0} \frac{dx}{e^x+e^{-x}} + \lim_{b\to\infty} \int_{0}^{b} \frac{dx}{e^x+e^{-x}} \quad [\text{choosing } c=0]$$

$$= \lim_{a\to-\infty} \left[\tan^{-1} e^x\right]_a^0 + \lim_{b\to\infty} \left[\tan^{-1} e^x\right]_0^b$$

$$= \lim_{a\to-\infty} (\tan^{-1} e^0 - \tan^{-1} e^a) + \lim_{b\to\infty} (\tan^{-1} e^b - \tan^{-1} e^0)$$

$$= \left(\frac{\pi}{4} - 0\right) + \left(\frac{\pi}{2} - \frac{\pi}{4}\right)$$

$$= \frac{\pi}{2}$$

(Answer)

Answer to the Question No - 3

My student is ~~is~~ : 24101226 , $a = 6$, $b = 2$

$A = 7$, $B = 3$

(a) $f(x) = -Ax + B$

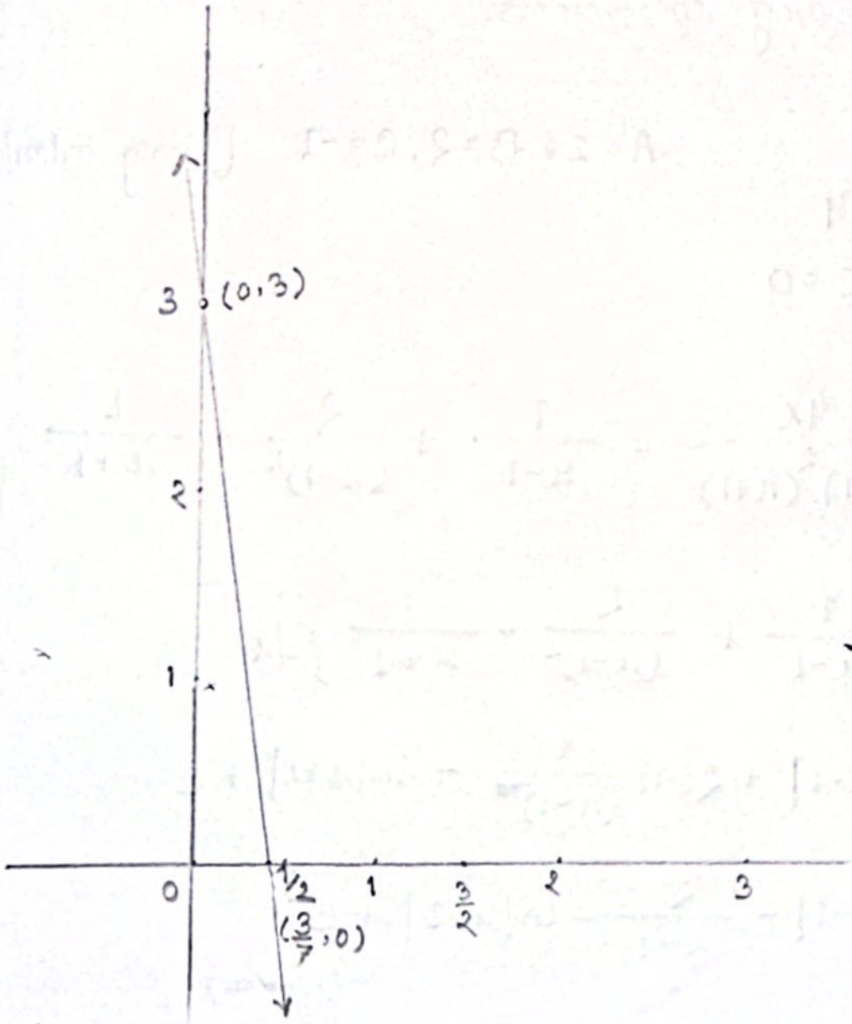
$\therefore f(x) = -7x + 3$

(b) If $f(x) = y = 0$

Interval $[\frac{1}{2}, \frac{3}{2}]$

$\Rightarrow -7x + 3 = 0$

$\Rightarrow x = \frac{3}{7} = 0.42857$



From the graph, it ~~is shown~~ shows that the range of interval is out of the range of the curve. So, area under the curve $f(x)$ over the ~~inter~~ interval $[\frac{1}{2}, \frac{3}{2}]$ is 0 square unit.

Answer to the Question No-4

$$f(x) = \frac{4x}{x^3-x^2-x+1}$$

[The function is a proper function as the degree of the numerator is lower than the degree of the denominator]

$$= \frac{4x}{(x-1)(x^2-1)}$$

$$f(x) = \frac{4x}{(x-1)^2(x+1)} \equiv \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$$

$$\Rightarrow 4x = Ax^2 - A + Bx + B + Cx^2 - 2Cx + C$$

$$\Rightarrow 0 \cdot x^2 + 4x + 0 = (A+C)x^2 + (B-2C)x + (-A+B+C)$$

Equating corresponding coefficients,

$A+C=0$

$B-2C=4$

$-A+B+C=0$

$A=1,\ B=2,\ C=-1$ [Using calculator]

$$f(x) = \frac{4x}{(x-1)^2(x+1)} = \frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{1}{x+1}$$

$$\int f(x)dx = \int \left\{\frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{1}{x+1}\right\}dx$$

$$= \ln|x-1| + 2(-1)\frac{1}{(x-1)} - \ln|x+1| + c$$

$$= \ln|x-1| - \frac{2}{x-1} - \ln|x+1| + c$$

(Answer)